Integration: Hadrian

Purpose
This interface is used to fetch Transportation values and Flex calculations to be used as part of the pricing calculations in the Pricing Tool but it is only required when generating certain types of offer

The Pricing Tool needs to call the Hadrian model (on internal ØRSTED server) in order to get calculated values when pricing some products.
The Hadrian model is subject to change as it gets developed internally.

Method
The communication between the Pricing Tool and Hadrian is done by sending a request XML file and receiving an XML response file over HTTP (Current message queue is not cloud based)

Below is a snippet of the Java code from the current solution which handles the communication with Hadrian today. It shows which inputs are provided to the Transport Model in Hadrian.

Hadrian then returns four numbers
- monthlyPremium
- averageYearlyCapacityPrice
- standaloneMonthlyPremium
- standaloneAverageYearlyCapacityPrice

However, the details and formats of this are less relevant in terms of a new Pricing Tool.
As long as the new Pricing Tool can provide the inputs listed below (numbers, strings, time series) via XML, to an endpoint that we control, then we can construct the endpoint such that Hadrian will receive the input it needs in the required format.

```
  /**
   * @param sourceId A string identifier.
   * @param countryCode A string country code.
   * @param notationDay Notationday. Actual used notation day will be nearest to or equal t
o this day.
   * @param deliveryPeriodStart First day of delivery.
   * @param deliveryPeriodEnd First day after last delivery day. If the delivery period id
Calendar year 2018,
   *     deliveryPeriodStart is 2018-01-01 and deliverPeriodEnd is 2019-01-01.
   * @param historicalPeriodStart First day of historical day.
   * @param historicalPeriodEnd First day after last day of historical off-
take. If the historical period
   *     id Calendar year 2018, historicalPeriodStart is 2018-01-
01 and historicalPeriodEnd is 2019-01-01.
   * @param historicalOfftake Time series with historical off-take (hourly granularity)
   * @param expectedOfftake Time series with expected load (Average daily load)
```

```java
   * @param yearlyCapacityPrice Yearly capacity price in DKK/hour/year/m3
   * @param quarterlyCapacityPrice Quarterly capacity price in DKK/hour/quarter/m3
   * @param monthlyCapacityPrice Monthly capacity price in DKK/hour/month/m3
   * @param dailyCapacityPrice Daily capacity price in DKK/hour/day/m3
   * @param addOn Add-on to the capacity price in DKK/hour/year/m3
   */
  public static String transportModel(String sourceId, String countryCode, GregorianCalenda
r notationDay,
      GregorianCalendar deliveryPeriodStart, GregorianCalendar deliveryPeriodEnd,
      GregorianCalendar historicalPeriodStart, GregorianCalendar historicalPeriodEnd,
      TimeSerieType historicalOfftake, TimeSerieType expectedOfftake,
      Double yearlyCapacityPrice, Double quarterlyCapacityPrice, Double monthlyCapacityPric
e,
      Double dailyCapacityPrice, Double addOn)
```